

Artificial Intelligence & Machine Learning – Task 2

Methodology, Results and Conclusion

Methodology

This project focused on improving a House Price Prediction system using the California Housing Dataset. The dataset was loaded and divided into input features and the target variable (House Price). Feature scaling was performed using StandardScaler to ensure that all features had a similar range, which improves model performance and stability. The dataset was then split into training and testing sets using an 80:20 ratio.

Three regression algorithms were trained and compared: Linear Regression Ridge Regression Decision Tree Regressor Model performance was evaluated using RMSE (Root Mean Squared Error) and R^2 Score. Lower RMSE indicates better prediction accuracy, while higher R^2 values indicate a better fit to the data.

Results

The trained models were evaluated on unseen test data. Feature scaling improved the learning process and helped create a fair comparison among models. The performance metrics were used to identify the most suitable model for house price prediction.

General observations: Linear Regression provided a strong baseline model. Ridge Regression reduced overfitting and improved generalization. Decision Tree Regression captured non-linear relationships in the housing data. An Actual vs Predicted visualization was also used to validate model performance. Models with points closer to the reference line produced more accurate predictions.

Conclusion

This project demonstrated the importance of data preprocessing, feature scaling, and model comparison in machine learning workflows. Instead of relying on a single algorithm, multiple models

were trained and evaluated to identify the best-performing approach. The use of RMSE and R^2 metrics provided an objective method for measuring model quality.

The project highlights a real-world machine learning pipeline where data preparation, model optimization, evaluation, and result interpretation are essential steps. The selected best-performing

model can be used as a foundation for future improvements and deployment in practical house price prediction applications.